PHYSICS

1) A block of metal weighing 3 kg is resting on a frictionless plane. It is struck by a jet releasing water at a rate of 6 kg/s with a speed of 4 m/s. Water does not rebound. The initial acceleration of the block will be :-

(1) 6 m/s^2

(2) 8 m/s^2

(3) 10 m/s^2

(4) 12 m/s^2

2) A force F is applied on block A as shown in figure. The contact force between the blocks A and B and between the blocks B and C respectively are (Assume frictionless surface) ?



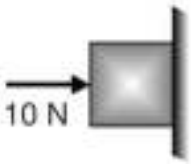
(1) $\frac{F}{7}, \frac{2F}{7}$

(2) $\frac{6F}{7}, \frac{4F}{7}$

(3) $F, \frac{F}{2}$

(4) $\frac{4F}{7}, \frac{6F}{7}$

3) A horizontal force of 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is



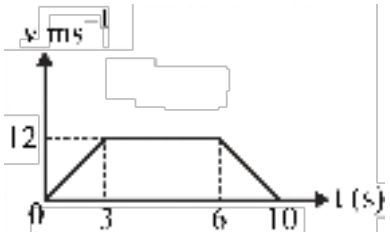
(1) 2 N

(2) 20 N

(3) 50 N

(4) 100 N

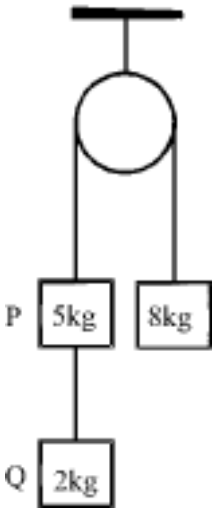
4) A lift is moving in upward direction. The total mass of the lift and the passenger is 1600 kg. The variation of the velocity of the lift is as shown in the figure. The tension in the rope at t = 8 seconds

will be 

(1) 11200 N
(2) 16000 N
(3) 48000 N
(4) 12000 N

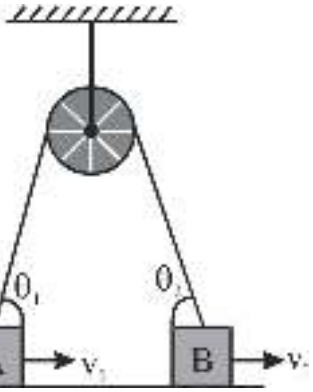
5) During the swinging of simple pendulum,

(1) the work done by tension force is always zero
(2) the work done by the gravitational force is zero
(3) the mechanical energy of the bob remains constant in the presence of air resistance
(4) the mechanical energy of the bob does not remain constant in the absence of air

6) In the following fig., two masses P and Q are joined with a string. The tension in the string in between PQ in newton will be- 

(1) g
(2) $\frac{g}{15}$
(3) $\frac{16g}{15}$
(4) $\frac{32g}{15}$

7) In Fig. blocks A and B move with velocities v_1 and v_2 along horizontal direction. Find the ratio of



v_1/v_2.

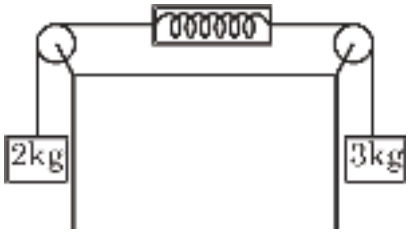
(1) $\frac{\sin\theta_1}{\sin\theta_2}$

(2) $\frac{\sin\theta_2}{\sin\theta_1}$

(3) $\frac{\cos\theta_2}{\cos\theta_1}$

(4) $\frac{\cos\theta_1}{\cos\theta_2}$

8) What will be the reading of spring balance?



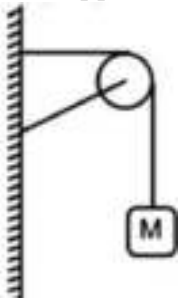
(1) 48 N

(2) 24 N

(3) 12 N

(4) 25 N

9) A massless string going over a clamped pulley of mass M supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given



(1) 2Mg

(2) $\sqrt{2}$Mg

(3) $\sqrt{3}$Mg

(4) $\sqrt{5}$Mg

10) A rope of length L is pulled by a constant force F. What is the tension in the rope at a distance x from the end where the force is applied :-

(1) $\frac{FL}{x}$

(2) $\frac{F(L-x)}{L}$

(3) $\frac{FL}{L-x}$

(4) $\frac{Fx}{L}$

11) A man pulls a bucket full of water from h metre deep well. If the mass of rope is m and mass of bucket full of water is M, then work done by the man is :-

(1) $\left(\frac{M}{2}+m\right)gh$

(2) $\left(\frac{M+m}{2}\right)gh$

(3) $\left(M+\frac{m}{2}\right)gh$

(4) (M + m) gh

12) Find acceleration of block. 

(1) 2 m/s^2

(2) 3 m/s^2

(3) 5 m/s^2

(4) 1 m/s^2

13) A bomb initially at rest explodes by itself into three equal mass fragments. The velocities of two fragments are $(3\hat{i}+2\hat{j})$ m/s and $(-\hat{i}-4\hat{j})$ m/s. The velocity of the third fragment (in m/s) is :-

(1) $2\hat{i}+2\hat{j}$

(2) $2\hat{i}-2\hat{j}$

(3) $-2\hat{i}+2\hat{j}$

(4) $-2\hat{i}-2\hat{j}$

14) A 2 kg body and a 3 kg body are moving along the x-axis. At a particular instant the 2 kg body has a velocity of 3 m/s and the 3 kg body has the velocity of 2 m/s. The velocity of the centre of mass at that instant is :-

(1) 5 m/s

(2) 1 m/s

(3) 0

(4) $\frac{12}{5}$m/s

15) Identify the wrong statement.

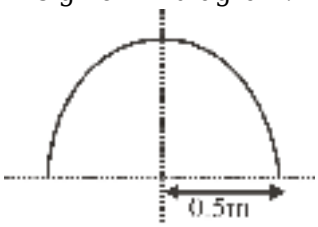
(1) A body can have momentum without mechanical energy

(2) A body can have energy without momentum

(3) The momentum is conserved in an elastic collision only.

(4) Kinetic energy is not conserved in an inelastic collision

16) A semicircular ring of radius 0.5m is given in diagram. The location of centre of mass on its line of symmetry above its base will be :-



(1) $\frac{2}{3\pi}$m

(2) $\frac{2}{\pi}$m

(3) $\frac{1}{\pi}$ m

(4) $\frac{3}{2\pi}$ m

17) A ball is dropped from a height h on the ground. If the coefficient of restitution is e, the height to which the ball goes up after it rebounds for n^{th} collision is :

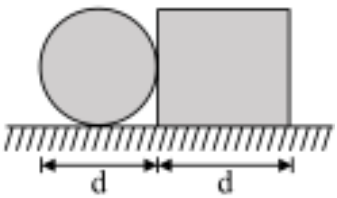
(1) he^{2n}

(2) he^{n}

(3) $\frac{e^{2n}}{h}$

(4) $\frac{h}{e^{2n}}$

18) A circular plate of diameter d is kept in contact with a square plate of edge d as shown in figure. The density of the material and the thickness are same everywhere. The centre of mass of the composite system will be



(1) inside the circular plate

(2) inside the square plate

(3) at the point of contact

(4) outside the system

19) A particle of mass m moving on a smooth surface with velocity 10m/s strikes another particle of mass 2m moving with 5 m/s in the same direction. If the collision is elastic and head-on, then velocities of particles in m/s after the collision are:-

(1) $\frac{10}{3}, \frac{25}{3}$

(2) $\frac{5}{3}, \frac{5}{3}$

(3) 5, 10

(4) 10, 5

20) A man A, mass 60 kg and other man B, mass 70 kg are sitting at two extremes of 2m long boat of mass 70 kg standing still in water as shown in fig. They come to the middle of boat (neglect friction). How far does the boat move on the water during the process ?



(1) 5 cm leftward

(2) 5 cm Rightward

(3) 7 cm leftward

(4) 7 cm Rightward

21) The kinetic energy of a body is increased by 21%. What is the percentage increase in the magnitude of linear momentum of the body :-

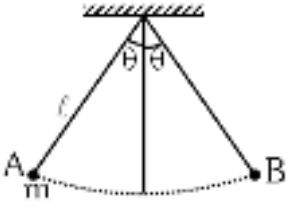
(1) 5%

(2) 10%

(3) 20%

(4) 21%

22) Find work done by tension when particle goes from A to B :-



(1) +ve

(2) –ve

(3) Zero

(4) None

23) A spring of spring constant 5×10^3 N/m is stretched initially by 5 cm from the unstretched position. Then the work required to stretch it further by another 5 cm is :-

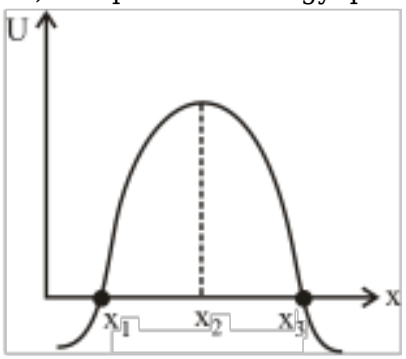
(1) 6.25 J

(2) 12.50 J

(3) 18.75 J

(4) 25.00 J

24) The potential energy- position graph of a particle is shown in figure. Then particle :-



(1) is in stable equilibrium at x_1

(2) is in stable equilibrium at x_2

(3) is in unstable equilibrium at x_2

(4) is in stable equilibrium at x_3

25) A particle moves with a velocity $\vec{v} = (5\hat{i} - 3\hat{j} + 6\hat{j})$ m/s under the influence of a constant force $\vec{F} = (10\hat{i} + 10\hat{j} + 20\hat{k})$N. The instantaneous power delivered to the particle is :-

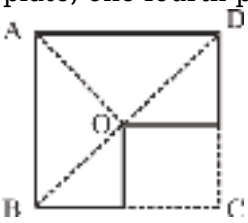
(1) 200 J/s

(2) 40 J/s

(3) 140 J/s

(4) 170 J/s

26) From a uniform square plate, one fourth part is removed as shown. The centre of mass of remaining part will be on ?



(1) OC

(2) OA

(3) OB

(4) OD

27) An engine pumps water continuously through a hose. Water leaves the hose with a velocity v and m is the mass per unit length of the water jet. What is the rate at which kinetic energy is imparted to water :-

(1) $\frac{1}{2} m^2v^2$

(2) $\frac{1}{2} mv^3$

(3) mv^3

(4) $\frac{1}{2} mv^2$

28)

Displacement of a body of mass 2 kg is given by $x = \frac{t^3}{3}$. Find work done in 2 seconds.

(1) 12 J

(2) 9 J

(3) 6 J

(4) 16 J

29) The Apparent coefficient of expansion of a liquid when heated in brass vessel is x and when heated in a tin vessel is y, if α is the coefficient of linear expansion for brass, the linear expansion of tin is:-

(1) $\frac{x + y + 3\alpha}{3}$

(2) $\frac{x + 3\alpha - y}{3}$

(3) $\frac{x + y - 2\alpha}{3}$

(4) $\frac{x + y - 2\alpha}{2}$

30) On heating a uniform metallic cylinder length increased by 3 %. The area of cross section of its base will be increased by :-

(1) 1.5 %

(2) 3 %

(3) 9 %

(4) 6 %

31) 50 gram of ice at 0°C is mixed with 50 gram of water at 60°C, final temperature of mixture will be :-

(1) 15°C

(2) 0°C

(3) 10°C

(4) 40°C

32) A beaker is completely filled with water at 4°C. It will overflow if

(1) heated above 4°C

(2) cooled below 4°C

(3) both (1) & (2)

(4) none of the above

33) On a 'X' temperature scale, water freezes at –125.0° X and boils at 375.0° X. On a 'Y' temperature scale, water freezes at –70.0° Y and boils at –30.0° Y. The value of temperature on X–scale equal to the temperature of 50.0°Y on Y–scale is

(1) 455.0°X

(2) –125.0°X

(3) 1375.0°X

(4) 1500.0°X

34) The co-efficient of linear expansion of crystal in one direction is α_1 and in other two directions it is α_2, then co-efficient of volume expansion of crystal will be :-

(1) $2\alpha_1 + \alpha_2$

(2) $3\alpha_1$

(3) $\alpha_1 + 2\alpha_2$

(4) $3\left(\frac{\alpha_1 + \alpha_2}{2}\right)$

35) An ice block at 0°C is dropped from height 'h' above the ground. What should be the value of 'h' so that it melts completely by the time it reaches the bottom assuming the loss of whole gravitational potential energy is used as heat by the ice ? [Given : $L_{ice} = 3.36 \times 10^5$ J/kg]

(1) 33.6 m

(2) 33.6 km

(3) 8 m

(4) 8 km

36) When a bimetallic strip is heated, it :-

(1) Does not bend at all

(2) Gets twisted in the form of an helix

(3) Bend in the form of an arc with the more expandable metal outside

(4) Bends in the form of an are with the more expandable metal inside

37) The temperature of a body on Kelvin scale is found to be x K. When it is measured by Fahrenheit thermometer, it is found to be x°F, then the value of x is :

(1) 40

(2) 313

(3) 574.25

(4) 301.25

38) A uniform metal rod is used as a bar pendulum. If the room temperature rises by 10°C, and the coefficient of linear expansion of the metal of the rod is 2×10^{-6} per °C, the period of the pendulum will have percentage increase of :-

(1) -2×10^{-3}

(2) -1×10^{-3}

(3) 2×10^{-3}

(4) 1×10^{-3}

39) A vessel contains 110 g of water. The heat capacity of the vessel is equal to 10g of water. The initial temperature of water in vessel is 10°C. If 220 g of hot water at 72°C is poured in the vessel, the final temperature neglecting radiation loss, will be :

(1) 70°C

(2) 80°C

(3) 60°C

(4) 50°C

40) Two rods, one made of aluminium and the other made of steel, having initial lengths $\square_1$ and $\square_2$ respectively are connected together to form a single rod of length ($\square_1 + \square_2$). The coefficients of linear expansion for aluminium and steel of α_1 and α_2 respectively. If the length of each rod increases by the same amount when their temperature is raised by t°C, then the ratio $\square_1/(\square_1 + \square_2)$:-

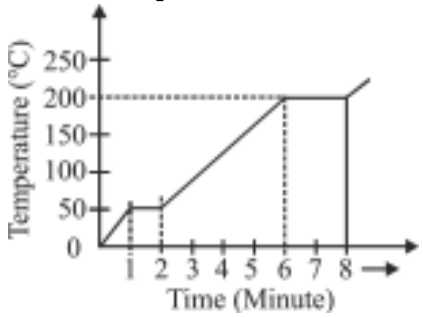
(1) $\frac{\alpha_1}{\alpha_2}$

(2) $\frac{\alpha_2}{\alpha_1}$

(3) $\frac{\alpha_2}{(\alpha_1 + \alpha_2)}$

(4) $\frac{\alpha_1}{(\alpha_1 + \alpha_2)}$

41) A student takes 50 gm wax (specific heat = 0.6 kcal/kg°C) and heats it till it boils. The graph between temperature and time is as follows. Melting point and boiling point of wax are respectively



:-

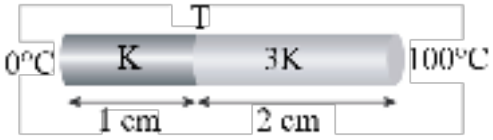
(1) 50°C, 50°C

(2) 50°C, 100°C

(3) 50°C, 200°C

(4) 50°C, 150°C

42) Two bars of thermal conductivities K and 3 K and lengths 1 cm and 2 cm respectively have same cross-sectional area. They are joined lengths wise as shown in the figure. If the temperature at the ends of this composite bar is 0°C and 100°C respectively (see figure), then the temperature of the interface (T) is :-



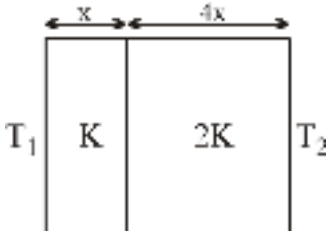
(1) 50°C

(2) $\frac{100}{3}$ °C

(3) 60°C

(4) $\frac{200}{3}$ °C

43) The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and 2K and thickness x and 4x respectively, are T_2 and T_1 ($T_2 > T_1$). The rate of heat transfer through the slab in a steady state is $\left(\frac{A(T_2 - T_1)K}{x}\right)$f, when f equal to



(1) 1

(2) 1 / 2

(3) 2 / 3

(4) 1 / 3

44) Consider a slab consisting of two different materials having equal thicknesses and thermal conductivities K and 2K, respectively.The equivalent thermal conductivity of the slab in parallel combination :–

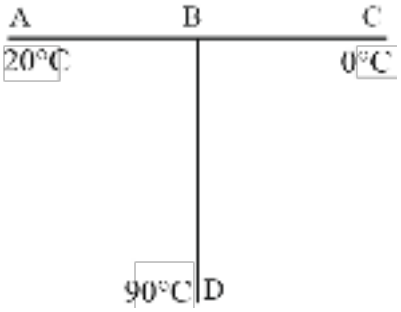
(1) $\frac{2}{3K}$

(2) $\sqrt{2}K$

(3) $\frac{3K}{2}$

(4) $\frac{4}{3K}$

45) Three conducting rods of same material and cross-section are shown in figure. Temperature of A, D and C are maintained at 20°C, 90°C and 0°C. Find the ratio of length BD and BC if there is no



heat flow in AB.

(1) $\frac{2}{7}$

(2) $\frac{1}{2}$

(3) $\frac{7}{2}$

(4) $\frac{1}{3}$

CHEMISTRY

1) **Formula** **Solubility product**

$AgBrO_3$ 2.5×10^{-10}

$MgSO_4$ 2.7×10^{-27}

$CaCO_3$ 2.7×10^{-18}

FeC_2O_4 2.4×10^{-15}

Which of the following salt having maximum solubility :-

(1) $AgBrO_3$

(2) $MgSO_4$

(3) $CaCO_3$

(4) FeC_2O_4

2) The number of hydroxide ions in 1 cc of a solution of pH = 12 is :

(1) 6.022×10^{7}

(2) 6.022×10^{22}

(3) 6.022×10^{19}

(4) 6.022×10^{18}

3) What will be the pH of the buffer solution containing 0.15 moles of NH_4OH & 0.25 moles of NH_4Cl (K_b for NH_4OH is 1.8×10^{-5})

(1) 4.96

(2) 10.04

(3) 8.04

(4) 9.04

4) Match the Column – I and Column - II

Column-I	Column-II
(A) $A\square(OH)_3$	(p) $S = \left(\frac{K_{sp}}{6912}\right)^{1/7}$
(B) $BaCrO_4$	(q) $S = \left(\frac{K_{sp}}{4}\right)^{1/3}$
(C) $Zr_3(PO_4)_4$	(r) $S = \left(\frac{K_{sp}}{27}\right)^{1/4}$
(D) Hg_2I_2	(s) $S = (K_{sp})^{1/2}$

(1) A → r;B → s;C → p;D → q

(2) A → s;B → p;C → r;D → q

(3) A → p;B → q;C → s;D → r

(4) A → q;B → r;C → p;D → s

5) The pH of the solution obtained by mixing 10mL of 10^{-1} N H_2SO_4 and 10mL of 10^{-1}N NaOH is :

(1) 8

(2) 2

(3) 7

(4) 1.3

6)

Aqueous solution		pH	
(A)	10^{-2} M HCl	(i)	12
(B)	10^{-2} M NaOH	(ii)	2
(C)	10^{-7} M HCl	(iii)	7 to 8
(D)	10^{-8} M NaOH	(iv)	6 to 7

(1) (A) – i, (B) –ii, (C) –iii, (D) -iv

(2) (A) – ii, (B) –iii, (C) –i, (D) -iv

(3) (A) – ii, (B) –i, (C) –iv, (D) -iii

(4) (A) – iv, (B) –iii, (C) –i, (D) -ii

7) Which of the following can not produce conjugate acid and conjugate base of the given species :

(1) H_2O

(2) HCO_3^-

(3) HSO_4^-

(4) NH_4^+

8)

The pH of blood circulating in a human body is maintained around 7.4 by the action of the buffer

system.

(1) CH_3COOH/CH_3COONa

(2) NH_4Cl/NH_3

(3) $H_2PO_4^{-2}$

(4) H_2CO_3/HCO_3^-

9) The solubility of $PbSO_4$ in 0.01 M Na_2SO_4 solution is (K_{sp} for $PbSO_4 = 1.25 \times 10^{-9}$) :-

(1) 1.25×10^{-7} mol L^{-1}

(2) 1.25×10^{-9} mol L^{-1}

(3) 1.25×10^{-10} mol L^{-1}

(4) 1.25×10^{-18} mol L^{-1}

10)

The ionization constant of CH_3COOH is 1.8×10^{-5}. The concentration at which it will be dissociated to 2% is :-

(1) 1 M

(2) 0.045 M

(3) 0.018 M

(4) 0.45 M

11) In the dissociation,



(1) K_1 is equal to K_2

(2) K_1 is smaller than K_2

(3) K_1 is greater than K_2

(4) K_1 is negligible

12) The pH of pure water is 6.63. This value indicates that the temperature of water is :-

(1) 298 K

(2) < 298 K

(3) > 298 K

(4) None of these

13) When enthalpy of formation of ammonia is –46 kJ/mol. The enthalpy change for the reaction $2NH_{3(g)} \rightarrow N_{2(g)} + 3H_{2(g)}$ will be :-

(1) 46 kJ mol^{-1}

(2) 92 kJ mol^{-1}

(3) –23 kJ mol^{-1}

(4) –92 kJ mol^{-1}

14) Based on the given values of bond energy :
N–N = 159 kJ mol^{-1}; H–H = 436 kJ mol^{-1}
N ≡ N = 941 kJ mol^{-1}; N–H = 398 kJ mol^{-1}
Heat of formation of $N_2H_4(g)$ (in kJ mol^{-1}) will be :-

(1) 711
(2) 62
(3) –98
(4) –711

15) The bond dissociation energies of X_2, Y_2 & XY are in the ratio of 1 : 0.5: 1. ΔH for the formation of XY is –200 kJ mol^{-1}. The bond dissociation energy of X_2 will be :-

(1) 800 kJ mol^{-1}
(2) 100 kJ mol^{-1}
(3) 200 kJ mol^{-1}
(4) 400 kJ mol^{-1}

16) The heat of combustion of C_2H_4, C_2H_6 and H_2 are –1409.5 kJ, –1558.3 kJ and –285.6 kJ. The heat of hydrogenation of ethene (C_2H_4) is :-

(1) –136.8 kJ
(2) –13.68 kJ
(3) 273.6 kJ
(4) 1.368 kJ

17) If $H_{2(g)} \rightarrow 2H_{(g)}$; ΔH = 104 kcal, then heat of atomisation of hydrogen (H_2) is :-

(1) 52 kcal
(2) 104 kcal
(3) 208 kcal
(4) 5.2 kcal

18) Which of the following reaction is showing equation corresponding to heat of formation as well as combustion ?

(1) $2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$
(2) $N_2(g) + O_2(g) \rightarrow 2NO(g)$
(3) $2NO_{(g)} \rightarrow N_2O_{4(g)}$
(4) $C_{(graphite)} + O_{2(g)} \rightarrow CO_{2(g)}$

19) Which of the following equation correctly represents the ΔH_f^o of ethane?

(1) 2C (diamond) + $3H_2(g) \rightarrow C_2H_6$ (g)

(2) $C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$

(3) $2C\ (graphite) + 3H_2(g) \rightarrow C_2H_6(g)$

(4) $4C\ (graphite) + 6H_2(g) \rightarrow 2C_2H_6(g)$

20) What amount of energy (kJ) is released in the combustion of 5.8 g of $C_4H_{10}(g)$?
$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(\square)$; $\Delta H° = -5756$ kJ

(1) 575.6

(2) 287.8

(3) 182

(4) 57.56

21) The heat of neutralization of HCl by NaOH is –55.9 kJ/mol. If the heat of neutralization of HCN by NaOH is –12.1 kJ/mol. The enthalpy of dissociation of HCN is

(1) – 43.8 kJ

(2) 43.8 kJ

(3) 68 kJ

(4) – 68 kJ

22) The equilibrium constant for a reaction is 10. What will be the value of $\Delta G°$ at 300 K ?

(1) –5.74 kJ

(2) –574 kJ

(3) + 11.48 kJ

(4) +5.74 kJ

23) The equilibrium concentration of the species in the reaction $A + B \rightleftharpoons C + D$ are 3, 5, 10 and 15 $molL^{-1}$ respectively at 300 K, the ΔG° for the reaction is (in cal) :-

(1) 13.81

(2) –1381.8

(3) –138.18

(4) 1391.6

24) Which of the following reaction is expected never to be spontaneous :-

(1) $2O_{3(g)} \rightarrow 3O_{2(g)}$ ΔH = –ve, ΔS = +ve

(2) $Mg_{(s)} + H_{2(g)} \rightarrow MgH_{2(s)}$ ΔH = –ve, ΔS = –ve

(3) $Br_{2(\ell)} \rightarrow Br_{2(g)}$ ΔH = +ve, ΔS = +ve

(4) $2Ag_{(s)} + 3N_{2(g)} \rightarrow 2AgN_{3(s)}$ ΔH = +ve, ΔS = –ve

25) For the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$,
the standard free energy $\Delta G° > 0$. The equilibrium constant (K) would be :–

(1) K = 0

(2) $K > 1$

(3) $K < 1$

(4) $K = 1$

26) Which of the following is not a rule for calculating oxidation number ?

(1) For ions, sum of oxidation number of all atoms is equal to the charge on the ion.

(2) The oxidation number of oxygen is –2 in all of its compounds.

(3) The oxidation number of fluorine is –1 in all of its compounds.

(4) Oxidation number of hydrogen is +1 except in binary hydrides of alkali metals and alkaline earth metals where it is –1.

27) What is the average oxidation number of sulphur in $Na_2S_4O_6$:-

(1) $\frac{+2}{3}$

(2) $\frac{+3}{2}$

(3) $\frac{+3}{5}$

(4) $\frac{+5}{2}$

28) The oxidation number of Oxygen in KO_3, Na_2O_2 is :-

(1) 3, 2

(2) 1, 0

(3) 0, 1

(4) – 0.33, –1

29) In the following reaction,
$As_2S_3 + H^+ + NO^{-3} \rightarrow NO + H_2O + AsO_4^{-3} + SO_4^{2-}$, the equivalent weight of As_2S_3 is related to its molecular weight by:-

(1) M/2

(2) M/4

(3) M/24

(4) M/28

30) A compound contains atoms of three elements A, B and C. If the oxidation number of A is +2, B is +5 and that of C is –2, the possible formula of the compound is :-

(1) $A_3(BC_4)_2$

(2) $A_3(B_4C)_2$

(3) ABC_2

(4) $A_2(BC_3)_2$

31) In the reaction $MnO_4^- + xH^+ + ne^- \rightarrow Mn^{+2} + yH_2O$. What is the value of n ?

(1) 5

(2) 8

(3) 6

(4) 3

32) In the redox reaction, $Zn + NO_3^- + H^+ \rightarrow 4Zn^{+2} + NH_4^+ + 3H_2O$
the correct coefficients for the reactants are :

	Zn	NO_3^-	H^+
(a)	2	1	5
(b)	5	3	8
(c)	4	1	10
(d)	3	2	5

(1) a

(2) b

(3) c

(4) d

33) In the following reaction what will be the ratio of coefficient of MnO_4^- and Fe^{+2} after balancing respectively $MnO_4^- + Fe^{+2} \rightarrow Mn^{+2} + Fe^{+3}$

(1) 1 : 2

(2) 1 : 3

(3) 1 : 4

(4) 1 : 5

34) Which of the following is disproportionation ?

(1) $MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$

(2) $Fe^{2+} + MnO_4^- + H^+ \rightarrow Fe^{3+} + Mn^{2+} + H_2O$

(3) $I_2 + OH^- \rightarrow I^- + IO^- + H_2O$

(4) $Cr^{3+} + IO_3^- + OH^- \rightarrow Cr_2O_7^{2-} + I^- + H_2O$

35)

Select the redox reaction ?

(1) $F^- + H^+ \rightarrow HF$

(2) $NaF \xrightarrow{\Delta} Na^+ + F^-$

(3) $H_2 + Cl_2 \rightarrow HCl$

(4) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

36)

Which of the following hydrides has lowest boiling point :-

(1) HF
(2) HCl
(3) HBr
(4) HI

37)

Which of the following molecules are attracted by Debye force :-

(1) Na^+ and HCl
(2) Xe and H_2O
(3) CO_2 and CO_2
(4) HBr and PCl_3

38) The lattice energy of sodium compounds is in following order :-

(1) NaF > NaCl > NaBr > NaI
(2) NaCl > NaF > NaBr > NaI
(3) NaCl > NaBr > NaF > NaI
(4) NaI > NaBr > NaCl > NaF

39) Ionic conductivity of hydrated ions are in order as :-

(1) Li > Na > K > Rb > Cs
(2) Li < Na < K > Rb > Cs
(3) Li > Na < K > Rb < Cs
(4) Li < Na < K < Rb < Cs

40)

The most covalent halide is :-

(1) MgF_2
(2) $MgCl_2$
(3) $MgBr_2$
(4) MgI_2

41)

Which of the following has the highest melting point :-

(1) $AlCl_3$
(2) LiCl
(3) NaCl
(4) $BeCl_2$

42)

Which of the following is most Lewis Acidic :-

(1) BeF_2
(2) $BeCl_2$
(3) $BeBr_2$
(4) BeI_2

43)

Which of the following molecule has 3c-4e$^-$ bond :-

(1) Cl_2O_6
(2) N_2O_5
(3) Be_2Cl_4
(4) B_2H_6

44) Which of the following molecules on reaction with water gives ammonia ?

(1) Al_4C_3
(2) Mg_3N_2
(3) Be_2C
(4) CaC_2

45) Which of the following oxides is acidic ?

(1) B_2O_3
(2) Al_2O_3
(3) Ga_2O_3
(4) In_2O_3

BIOLOGY

1) Which is not macromolecule ?

(1) Protein
(2) Inulin

(3) Cellulose
(4) Glycerol

2) Which of the following is a secondary metabolite?

(1) Glucose
(2) Amino acid
(3) Rubber
(4) both 1 & 2

3) Cholesterol is :-

(1) Lipid
(2) Protein
(3) Alcoholic amino acid
(4) Acidic amino acid

4) Which of the following lipid is present in cell membrane ?

(1) Valine
(2) Lysine
(3) Lecithin
(4) Arachidonic acid

5) $CH_3 — (CH_2)_{14} — COOH$ is :-

(1) Amino acid
(2) Fatty acid [palmitic acid]
(3) Polysaccharide
(4) Monosaccharide

6) Which of the following is essential fatty acid ?

(1) Lauric acid
(2) Tryptophan
(3) Arachidonic acid
(4) Arginine

7) How many double bonds will be present in Linoleic acid between C-C atoms :-

(1) 1
(2) 2
(3) 3
(4) 4

8) Which of the following is reducing sugar ?

(1) Trehalose
(2) Sucrose
(3) Dihydroxy acetone
(4) Glycogen

9) Lecithin is :-

(1) Glycolipid
(2) Phospholipid
(3) Simple lipid
(4) Derived lipid

10) Which of the following is heteropolymer ?

(1) Protein
(2) Pectin
(3) Cellulose
(4) Both (1) and (2)

11) Fructose is -

(1) Laevorotatory
(2) Aldose
(3) Disaccharide
(4) Pentose

12) Dihydroxy acetone has -

(1) No chiral carbon
(2) Two hydroxyl groups
(3) No. of total carbons = 4
(4) Both 1 & 2

13)

Match the following :-

	Column-I		**Column-II**
a	Alkaloid	I	Vinblastin, curcumin
b	Essential oils	II	Morphine, Codeine
c	Toxins	III	Lemon grass oil
d	Drugs	IV	Abrin, Ricin

(1) a-II, b-III, c-IV, d-I

(2) a-III, b-II, c-IV, d-I
(3) a-II, b-III, c-I, d-IV
(4) a-III, b-II, c-I, d-IV

14) Select the odd one w.r.t. primary metabolites.

(1) Amino acids
(2) Sugars
(3) Drugs
(4) Fatty acid

15) In proteins, amino acids are attached together by :

(1) Peptide bond
(2) Amide bond
(3) Ester bond
(4) 1 & 2 both

16) α-amino acids are those, which –

(1) Participates in protein synthesis
(2) –COOH group and $–NH_2$ group are attached on same carbon
(3) 20 types
(4) All the above are correct

17) Which of the following is aromatic AA?

(1) Tyrosine
(2) Tryptophan
(3) Phenyl alanine
(4) All the above

18)

Match the columns I and II, and choose the correct combination from the options given :-

Column-I		**Column-II**	
a.	Acidic amino acid	i.	Valine
b.	Basic amino acid	ii.	Glutamic acid
c.	Neutral amino acid	iii.	Phenylalanine
d.	Aromatic amino acid	iv.	Lysine

(1) a-ii, b-iv, c-i, d-iii
(2) a-ii, b-i, c-iv, d-iii
(3) a-iii, b-ii, c-i, d-iv

(4) a-i, b-iv, c-iii, d-ii

19) Two unit of glucose linked by α 1→ 4 glycosidic linkage in which type of Disaccharide

(1) Trehalose
(2) Maltose
(3) Lactose
(4) Mannose

20) Which of the following group is odd ?

(1) Starch, glycogen, cellulose
(2) Pectin, Dextrin, Inulin
(3) Heparin, Pectin, Chondriotin
(4) Hemicellulose, Chondriotin, Pectin

21) Which of the follwing pair is not correctly matched ?

(1) Ribose –$C_5H_{10}O_5$
(2) Mannose–$C_6H_{12}O_6$
(3) Glucose–$C_6H_{12}O_6$
(4) Erythrose–$C_3H_6O_3$

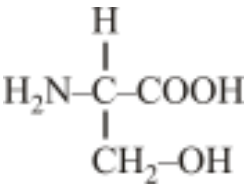
22) Find out the incorrect match ?

(1) Inulin - Polymer of fructose
(2) Starch - Spiral secondary structure
(3) Cellulose - Component of cell wall
(4) Glycogen - Monosaccharide and reserved food of plants

23) How many H and O atoms are present in deoxyribose sugar respectively ?

(1) 5, 10
(2) 4, 10
(3) 10, 5
(4) 10, 4

24) Following structure is related to which amino acid?



(1) Serine
(2) Glutamic acid
(3) Tyrosine

(4) Histidine

25)

Match the column.

	A		B
(A)	Maltose	(i)	Polysaccharide
(B)	Chitin	(ii)	Aromatic amino acid
(C)	Serine	(iii)	Disaccharide
(D)	Tryptophan	(iv)	Alcoholic amino acid

(1) A - iii ; B - i ; C - ii ; D - iv

(2) A - iii ; B - i ; C - iv ; D - ii

(3) A - i ; B - iii ; C - ii ; D - iv

(4) A - iii ; B - ii ; C - i ; D - iv

26) Which glycosidic linkage is found at branching point in glycogen ?

(1) β 1 → 4

(2) α 1 → 4

(3) α 1 → 6

(4) Both α 1 → 4 and α 1 → 6

27)

Which of the following amino acid is basic ?

(1) Valine

(2) Arginine

(3) Glutamic acid

(4) Glycine

28) Maltose is a :-

(1) Disaccharide

(2) Reducing sugar

(3) Malt sugar

(4) All

29) Exoskeleton of Arthro Pods and Fungal Cellulose contain ?

(1) N Acetyl Galactosamine

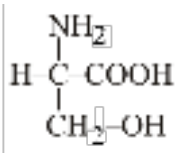
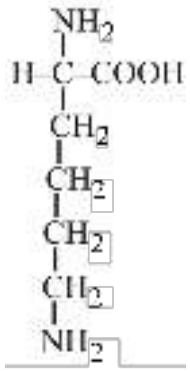
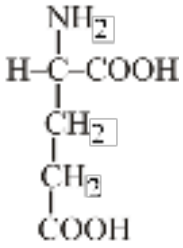
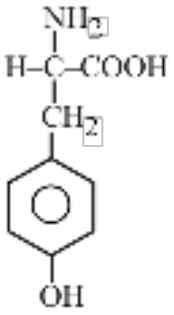
(2) N Acetyl Glucosamine

(3) N Acetyl Fructo samine

(4) β D Glucose

30)

Which one out of A-D given below correctly represents the structural formula of the acidic amino acid :-

A	B	C	D
			

(1) D

(2) C

(3) A

(4) B

31) Which one is the most abundant protein in the animal world ?

(1) Collagen

(2) Insulin

(3) Trypsin

(4) Haemoglobin

32) **Assertion :-** Glycine is neutral Amino acid.
Reason :- It contains one carboxylic group and one Amino group.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

33) Read the following statements :-
(a) Water is the most abundant chemical in living organisms.
(b) In a polysaccharide the individual monosaccharides are linked by a glycosidic bond.
(c) Lipids are generally water insoluble & many lipids have both glycerol and fatty acids.
(d) Protein is a heteropolymer and not a homopolymer.
How many statement is/are correct ?

(1) One

(2) Two

(3) Three

(4) Four

34) In alcoholic fermentation, the final e^- acceptor is :-

(1) Acetaldehyde
(2) Ethyl alcohol
(3) NADH + H^+
(4) Pyruvic acid

35) Pyruvic acid converted into ethyl alcohol by -

(1) Isomerase
(2) Alcohol dehydrogenase
(3) Hexokinase
(4) Both (1) & (3)

36) The metabolic breakdown of carbohydrates, fatty acids and amino acids is connected through

(1) Acetyl CoA
(2) Pyruvic acid
(3) PGAL
(4) α-ketoglutarate

37) Respiration is :-

(1) Amphibolic process
(2) Exergonic
(3) Downhill process
(4) All of the above

38) How many molecules of ATP are produced per molecule of $FADH_2$ oxidised ?

(1) One
(2) Three
(3) Four
(4) Two

39) Which of the following complexes are involved in $NADH_2$ and $FADH_2$ oxidation through ETS respectively :-

(1) I, II
(2) II, IV
(3) II, III
(4) III, V

40) Number of oxidation and decarboxylation steps in one TCA cycle is ___ and ___ respectively :-

(1) 4, 2

(2) 3, 1

(3) 2, 2

(4) 4, 3

41) Non cyclic photophosphorylation involved in the process of :

(1) ATP synthesis

(2) Non cyclic flow of electrons

(3) Formation of NADPH + H^+

(4) All of the above

42) For most of plants, light saturation occurs at___ of the full sunlight.
Fill in the blank with correct option.

(1) 50%

(2) 10%

(3) 90%

(4) 1%

43) Match Column-I with Column-II and select the correct answer using the codes given below :

Column-I		**Column-II**	
A.	4C Compound	I.	Acetyl CoA
B.	2C Compound	II.	Pyruvate
C.	5C Compound	III.	Citric acid
D.	3C Compound	IV.	α-ketoglutaric acid
		V.	Malic acid

(1) A-II; B-V; C-III; D-I

(2) A-V; B-I; C-IV; D-II

(3) A-III; B-I; C-IV; D-II

(4) A-V; B-III; C-I; D-II

44) The primary CO_2 acceptor in C_4 pathway is a :

(1) 5-C compound

(2) 3-C compound

(3) 4-C compound

(4) 6-C compound

45) The law of limiting factor for photosynthesis was given by :–

(1) R. Hill

(2) Krebs

(3) Calvin
(4) Blackman

46) In Kranz anatomy, the bundle sheath cells have:

(1) thick walls, many intercellular spaces and few chloroplasts
(2) thin walls, many intercellular spaces and no chloroplasts.
(3) thick walls, no intercellular spaces and large number of chloroplasts.
(4) thin walls, no intercellular spaces and no chloroplasts.

47) Which enzyme is absent in mesophyll cells of C_4 plants ?

(1) Rubisco
(2) PEPcase
(3) Both (1) and (2)
(4) None of the above

48) The primary acceptor of CO_2 in C_3 plants is :-

(1) RuBP
(2) PGA
(3) PEP
(4) Malic acid

49) PS I occurs in :-

(1) appressed part of granal thylakoids
(2) Appressed and non appressed part of grana thylakoids
(3) Stroma
(4) Stroma thylakoids and non-appressed part of grana thylakoids.

50) Chlorophyll contains :

(1) Fe
(2) Mg
(3) K
(4) Mn

51) The site for dark reaction of photosynthesis is :-

(1) Stroma
(2) Grana
(3) Intergrana
(4) Mitochondria

52) Photosynthetic photosystem consists of :-

(1) Reaction centre only
(2) LHC only
(3) Both LHC & reaction centre
(4) LHC, reaction centre and ETC

53) Which one of the following is precursor of protochlorophyll?

(1) Acetyl COA
(2) Succinyl COA
(3) Oxlaoacetic acid
(4) α-ketoglutarate

54) Which of the following is not always required for chemiosmosis ?

(1) Membrane
(2) Proton pump
(3) OEC
(4) ATP synthase

55) How does PS-II supply electrons continuously ?

(1) by removing electrons from photon
(2) by removing electrons from H_2O
(3) by removing electrons from CO_2
(4) by removing electrons from constituent carotenoids

56) Which of the following is not correct ?

(1) Light is rarely a limiting factor for photosynthesis
(2) Current availability of atmospheric CO_2 levels is not limiting for C_3 plants
(3) Law of limiting was given by Blackman
(4) Water stress indirectly reduces the CO_2 availability for plants

57) R.Q. is less than one at the time of respiration of :-

(1) Starch
(2) Fructose
(3) Glucose
(4) Fat

58) Which of the following is 5-carbon compound of Kreb's cycle ?

(1) Citric Acid
(2) α-ketoglutarate
(3) Fumarate

(4) Malate

59) The incomplete breakdown of sugars in anaerobic respiration results in the formation of

(1) Fructose and water
(2) Glucose and carbon dioxide
(3) Alcohol and CO_2
(4) Water and CO_2

60) The organism in which Kreb's cycle does not occur in mitochondria is

(1) Yeast
(2) *E.coli*
(3) *Ulothrix*
(4) Plants

61) Which of the following ETC complex is directly involved in reduction of oxygen ?

(1) complex-I
(2) complex-II
(3) complex-III
(4) complex-IV

62)

Which is the site of Kreb's cycle ?

(1) Chloroplast
(2) Golgibody
(3) Mitochondria
(4) Endoplasmic reticulum

63) Sporophyte of pteridophyte is :-

(1) Depend upon Gametophyte
(2) Free living, Independent
(3) Semiparasite upon Gametophyte
(4) Non-photosynthetic

64) Which of the following groups of plants have same ploidy of endosperm ?

(1) *Cycas*, Mango, *Cedrus*
(2) Pea, *Castor*, *Sequoia*
(3) *Pinus*, *Cycas*, *Cedrus*
(4) *Sequoia*, Maize, *Pteridium*

65) The independent gametophytic generations are found in :-

(1) *Cedrus* and *Funeria*

(2) *Gnetum* and *Polytrichum*

(3) *Ephedra* and *Sphagnum*

(4) *Marchantia* and *Adiantum*

66) In brown algae asexual spores are :-

(1) Pear-shaped and have two unequal flagella

(2) Pear-shaped and have two unequal cilia

(3) Oval-shaped and have two unequal flagella

(4) Comma-shaped and biflagellate

67) In which plants male gametes are motile :-
(A) Bryophyta (B) Pteridophyta
(C) Angiosperm (D) Algae

(1) A & B

(2) B & C

(3) A, B & C

(4) A, B & D

68) Which type of sexual reproduction is found in *Volvox* ?

(1) Isogamous

(2) Anisogamous

(3) Oogamous

(4) All of these

69) Pyrenoids are located in :-

(1) Nucleus

(2) Nucleolus

(3) Mitochondria

(4) Chloroplast

70) Gemmae are :-

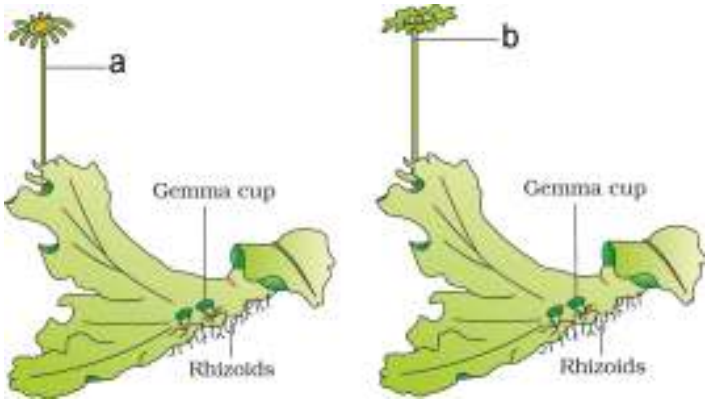
(1) Asexual reproductive structure of *cycas*

(2) Sexual reproductive structure of fern.

(3) Sexual reproductive structure of Moss

(4) Asexual resproductive structure of marchantia.

71) Find out the correct matching regarding a and b :-



(1) a-archegonia, b-antheridia

(2) a-antheridia, b-archegonia

(3) a-antheridiophore, b-archegoniophore

(4) a-archegoniophore, b-antheridiophore

72) Which of the following pteridophytes are heterosporous in nature ?

(1) *Selaginella* and *Salvinia*

(2) *Adiantum* and *Equisetum*

(3) *Psilotum* and *Lycopodium*

(4) *Adiantum* and *Psilotum*

73) Vascular tissues are found in _____.

(1) Algae

(2) Bryophytes

(3) Fungi

(4) Pteridophytes

74) Match the Kingdoms listed under column I with the characteristics give under column II. Choose the answer which give the correct combination :

Column I Kingdom		**Column II characteristics**	
(A)	Animalia	(P)	Unicellular prokaryotes
(B)	Plantae	(Q)	Multicellular heterotrophs
(C)	Protista	(R)	Multicellular autotrophs
(D)	Monera	(S)	Unicellular eukaryotes

(1) A–S, B–Q, C–R, D–P

(2) A–Q, B–R, C–S, D–P

(3) A–S, B–R, C–Q, D–P

(4) A–R, B–Q, C–S, D–P

75) Consider the following statements:-

(a) They grow in soil, on logs and tree stumps and in living plant bodies as parasites. (b) The mycelium is branched and septate and the asexual spores are generally not found but vegetative reproduction by fragmentation is common. (c) The sex organs are absent but plasmogamy is brought about by fusion of two vegetative or somatic cells of different strains or genotypes. Above statements are related to the member of which class :-

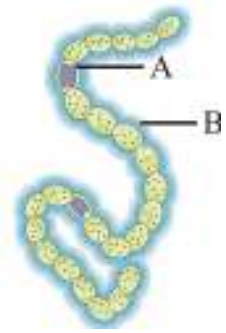
(1) Phycomycetes
(2) Ascomycetes
(3) Basidiomycetes
(4) Deuteromycetes

76) Which of the following kind of bacteria is considered to be most abundant in nature ?

(1) Chemosynthetic autotroph
(2) Heterotrophic
(3) Photosynthetic
(4) Nitrogen fixing

77) Methane can be produced by:-

(1) Acetyltroph
(2) Methanogen
(3) Thermoacidophile
(4) Halophiles



78) Identify correct option for above figure :-

	Structure	Function
(1)	A is heterocyst	Food Storage
(2)	A is heterocyst	N_2 fixation
(3)	B is mucilage sheath	Reproduction
(4)	B is filament	to make new sheath

(1) 1
(2) 2
(3) 3
(4) 4

79) Instead of cell wall, euglenoids have :-

(1) A protein rich slime layer
(2) A protein rich pellicle layer
(3) A lipid rich pellicle layer
(4) A lipo-protein glycocalyx

80) Choose the **incorrect** pair of following

(1) Phycomycetes — aquatic habitat
(2) Ascomycetes — Rust and smut disease
(3) Basidiomycetes — lack sex organs
(4) Deuteromycetes — help in mineral cycling

81) Which of the following is used extensively in biochemical and genetical work ?

(1) *Agaricus*
(2) *Alternaria*
(3) *Neurospora*
(4) *Mucor*

82) *Morchella* (Morels) is a member of class :-

(1) Phycomycetes
(2) Ascomycetes
(3) Deuteromycetes
(4) Basidiomycetes

83) Protein coat of virus is termed as :-

(1) Capsule
(2) Capsid
(3) Envelope
(4) Glycocalyx

84) Bacteria is a group of prokaryotic organisms which is characterized by :-

(1) 70s ribosomes
(2) Peptidoglycan cell wall
(3) ds circular naked DNA
(4) All of these

85) "Systema Naturae" was written by :-

(1) Carolus Linnaeus
(2) Theophrastus
(3) Bentham & Hooker
(4) John Ray

86) About how many species are known and described ?

(1) 30 million
(2) 1.7 to 1.8 million
(3) 0.2 million
(4) 17 million

87)

Which of the following is the most crucial step of the Calvin cycle ?

(1) Triose phosphate → Sucrose
(2) 3 - phosphoglycerate → Triose phosphate
(3) Ribulose-1, 5-bisphosphate + CO_2 → 3-phosphoglycerate
(4) Triose phosphate → starch

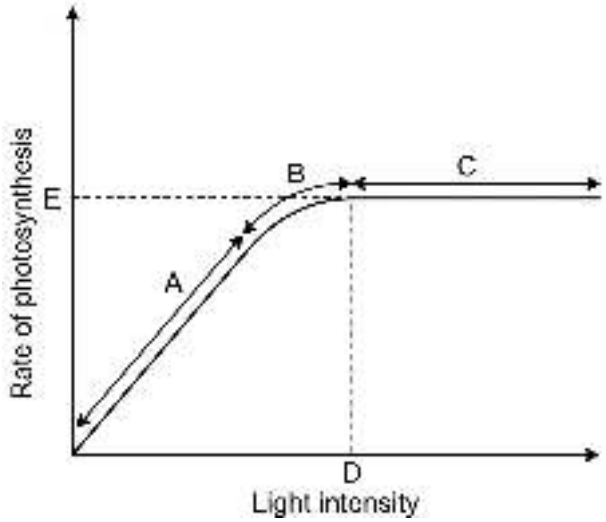
88) In Calvin cycle, CO_2 is fused with primary acceptor molecule to form 3C compound phosphoglycerate. If in this reaction CO_2 provides one carbon then how many carbons are present in primary acceptor molecule ?

(1) Five carbons
(2) Two carbons
(3) Six carbons
(4) Three carbons

89) RuBisCO has :-

(1) a greater affinity for O_2 than for CO_2
(2) a greater affinity for CO_2 than for O_2
(3) similar affinity for CO_2 and O_2
(4) affinity neither for O_2 nor for CO_2

90) Figure shows the effect of light on the rate of photosynthesis based on the figure select the correct option



(I) A - represent condition when light is

limiting.
(II) D - represents light saturation point.
(III) C - represents condition when light is not limiting.

(1) I and II only

(2) II and III only

(3) I and III only

(4) I, II and III

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	1	1	1	4	3	2	4	2	3	3	3	4	3	3	1	2	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	3	3	3	2	2	4	2	4	2	3	3	3	2	3	3	4	4	3
Q.	41	42	43	44	45															
A.	3	3	4	3	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	4	4	1	3	3	4	4	1	2	3	3	2	2	1	1	2	4	3	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	1	2	4	3	2	4	4	4	1	1	3	4	3	3	2	2	1	4	4
Q.	86	87	88	89	90															
A.	3	4	3	2	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	3	1	3	2	3	2	3	2	4	1	4	1	3	4	4	4	1	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	4	4	1	2	3	2	4	2	2	1	1	4	1	2	1	4	4	1	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	2	2	4	3	1	1	4	2	1	3	2	3	2	2	4	2	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	2	3	4	1	4	3	4	4	4	1	4	2	3	2	2	2	2	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	2	4	1	2	3	1	2	4										

SOLUTIONS

PHYSICS

1) $f = V\frac{dm}{dt}$

$ma = V\frac{dm}{dt}$

$3(a) = 4 \times 6$

$a = 8\ m/s^2$

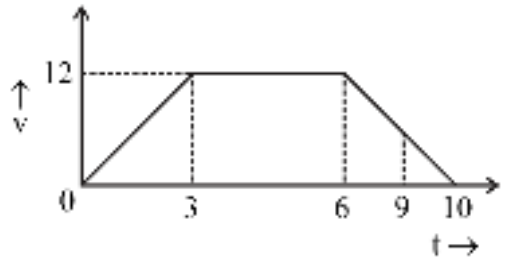
2) $a = \frac{F}{m + 2m + 4m} = \frac{F}{7m}$

$N_{AB} = (2m + 4m)a = \frac{6F}{7}$

$N_B = 4ma = \frac{4F}{7}$

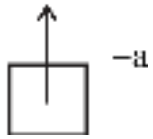
3) Given horizontal force $F = 10$ N and coefficient of friction between block and wall $(\mu) = 0.2$.
We know that at equilibrium the horizontal force provides the normal reaction to the block against the wall. Therefore, the normal reaction to the block $(R) = F = 10$ N.
We also know that weight of the block (W) = Frictional force $= \mu R = 0.2 \times 10 = 2$ N.

4)



at 8th second, the object is retarding upwards.

So,



$N = mg - ma$

$N = 1600(10) - 1600(3) \Rightarrow 1600(7)$

$N = 11200$ N

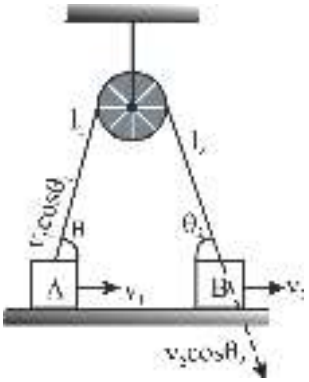
5) Since tension force is always perpendicular to the direction of instantaneous displacement

of the bob, the work done due to tension force is always zero.

6) $a = \frac{8g-(2+5)g}{8+2+5} = \frac{g}{15} = \frac{10}{15} m/s^2$

$T = 2(g + a) = 2\left(10 + \frac{10}{15}\right) = \frac{320}{15} = \frac{32g}{15}$

7) From Fig. $l_1 + l_2 = C$ or $\frac{dl_1}{dt} + \frac{dl_2}{dt} = 0$



$-v_1 \cos\theta_1 + v_2 \cos\theta_2 = 0$ or

8) $T = \frac{2m_1m_2}{m_1+m_2} \cdot g = 24N$ → Reading of spring

9) from theory

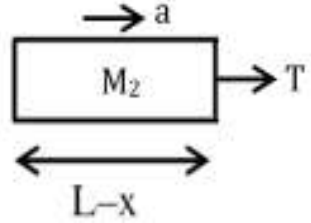
10)

Question Explanation : Rope is pulled by force question asks for tension in a rope at a distance x from application of force.

Concept : This question is based on newton's second law

Solution :

Acceleration of rope $a = \frac{F}{M}$



Mass (M_2) of rope of length (L–x) $= \frac{M}{L}(L-x)$

$T = M_2a$

$T = \frac{M}{L}(L-x)\frac{F}{M}$

$$T = \frac{F(L-x)}{L}$$

Final Answer : Option (2)

11) **Explain Question:**
The question asks for the total work done by a man in pulling a bucket full of water from a well of depth h.

Concept:
Work done by external force = Change in potential Energy.

Solution: WD by the man = change in P.E.
CMB = com of Bucket CMR = com of Rope W.D = mgh_{CMB} + mg h_{cmR}.

$$= mgh + mg\frac{h}{2}$$

$$W.D = mgh + mg\frac{h}{2}$$ option (3) correct

Final Answer: Option (3)

12) $a = \frac{100 - 0.5 \times 10g}{10}$

$= \frac{50}{10} = 5\ m/s^2$

13) **Solution:**
All three fragments have equal masses, say 'm' each,
Then, by COLM

$$m\vec{v}_3 + m\vec{v}_2 + m\vec{v}_1 = 0$$

$$\Rightarrow \vec{v}_3 = -(\vec{v}_2 + \vec{v}_1)$$

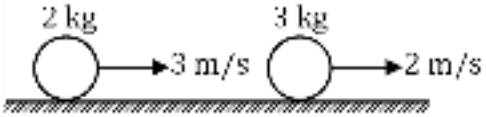
$$\vec{v}_1 = 3\hat{i} + 2\hat{j},\ \ \vec{v}_2 = -\hat{1} - 4\hat{j}$$

$$\vec{v}_3 = -\left[-\hat{i} - 4\hat{j} + 3\hat{i} + 2\hat{j}\right]$$

$$\Rightarrow \vec{v}_3 = -2\hat{i} + 2\hat{j}$$

Final Answer: Option (3)

14) **Explain Question:**
Find out velocity of the cm for two bodies moving along the x-axis.
Concept:
This Question is based on
Velocity of center of mass.
Solution:



$$\vec{v}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$$

$m_1 = 2kg \quad m_2 = 3kg$

$|\vec{v}_1| = 3m/s \quad |\vec{v}_2| = 2m/s$

$$= \frac{2 \times 3 + 3 \times 2}{2+3} = \frac{12}{5} \text{ m/s}$$

Final Answer:

Option (4)

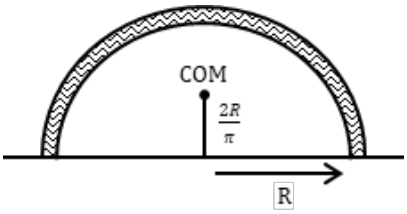
15) from theory

16)

Question Explanation: Find position of center of mass of a semicircular ring form its base.

Concept: Centre of mass.

Solution:



Centre of mass a semicircular ring lies at height $\frac{2R}{\pi}$ form its base.

$$r_{com} = \frac{2R}{\pi}$$

Given that R = 0.5 m

$$r_{com} = \frac{2 \times 0.5}{\pi} = \frac{1}{\pi} m$$

Final answer: option (3)

17)

After each collision speed becomes e times speed after 1st collison is $v_1 = ev_0$

$$h1 = \frac{v_1^2}{2g} = e^2 \left(\frac{v_0^2}{2g} \right) = e^2 h$$

Similarly maximum height after 2nd collision is,

$h_3 = e_2 h_2 = e^4 h$

Maxiumu height after 3nd collision is,

$h_3 = e^2 h_2 = e^6 h$

So, the height to which the ball goes up after it rebounds for then n^{th} time is $h_n = e^{2n} h$

18) **Explain Question:** We have to calculate centre of mass of combination.

Concept: Centre of mass of composite bodies

Solution:

Taking center of the disc as origin,

X coordinate of the disc, $x_1 = 0$

X coordinate of plate, $x_2 = \frac{a}{2} + \frac{a}{2} = a$

Centre of mass of the combination,

$$x_{cm} = \frac{A_1x_1 + A_2x_2}{A_1 + A_2}$$

$$x_{cm} = \frac{0 + a^2a}{\frac{\pi a^2}{4} + a^2} = \frac{4a}{4 + \pi} = 0.56\,a$$

Here, $x_{COM} > \frac{a}{2}$

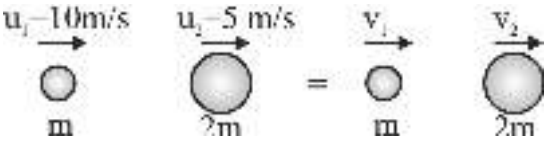
So, it lies inside the square plate

Final answer: Option (2)

19) **Explaining :** Find velocity of particles after collision.

Concept : Head on elastic collision

Solution :



By the momentum conservation

$$m \times 10 + 2m \times 5 = mv_1 + 2mv_2 \qquad \text{(i)}$$

We have another condition for elastic head-on collision

$$v_1 + u_1 = v_2 + u_2$$

$$v_1 + 10 = v_2 + 5 \qquad \text{(ii)}$$

From (i)

$$v_1 + 2v_2 = 20 \qquad \text{(iii)}$$

From (ii)

$$v_1 - v_2 = -5 \qquad \text{(iv)}$$

Solving, $v_1 = \frac{10}{3}$ m/s, $v_2 = \frac{25}{3}$ m/s

Final Answer : (1)

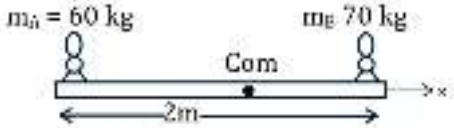
20)

Explain Question : Find shift of boat when man A and B cone to the middle of the boat.

Concept : Conservation of momentum (Men – plank problem)

Solution :

Initial position:



There is no external force C.O.M. remain of rest

$\Sigma m\Delta x = 0$

displacement of A (60kg) = 1 + x (Right)

displacement of B (70kg) = 1 – x (Left)

displacement of boat (70kg) = x (Right)

So, 70x + 60(1 + x) + 70 [–(1–x)] = 0
70x + 60 + 60x + 70x – 70 = 0
200x – 10 = 0
$x = \frac{10}{200}$ = 0.05 m = 5 cm.

Final Answer : option (2)

21)

Explain Question : To find percentage increases in magnitude of linear Momentum.
Gives – Parentage increased in kinetic energy = 21%

Solution :

$$K.E_f = \frac{121}{10}K.E_i \Rightarrow P_f = \frac{11}{10}P_i$$
$$\Delta P = \frac{1}{10}P_i$$
$$\frac{\Delta P}{P_i} = \frac{1}{10}$$
$$\frac{\Delta P}{P_i} \times 100 = 10\%$$

Final Answer : option (2)

22) **Explain Question :** As shown in figure find work done by tension.
Concept : This question is based on work done.
Explanation :
W = F.S cos θ
∴ θ = 90°
 Angle between force and displacement is always 90°
 cos 90° = 0
W = F.S. × 0
W = 0
Final answer : 3

23) $W = \frac{1}{2}k(x_2^2 - x_1^2)$
$$= \frac{1}{2} \times 5 \times 10^3 \times \left[\left(\frac{5+5}{100}\right)^2 - \left(\frac{5}{100}\right)^2\right]$$
⇒ W = 18.75 J

24) $F = -\frac{dU}{dx} = 0$ for equilibrium
and $\frac{d^2U}{dx^2} > 0$ for minimum potential energy and stable equilibrium.
$\frac{d^2U}{dx^2} < 0$ for maximum potential energy and unstable equilibrium.

25)

$P = \vec{F}.\vec{v} = 50 - 30 + 120 = 140\ J/s$

26)

COM lies on the axis of symmetry and closer to heavier mass.

27)

Here $m = \frac{dM}{dl}$ So, Rate of KE = $\frac{d(KE)}{dt}$

$\frac{d(KE)}{dt} = \frac{1}{2}\frac{dM}{dt}v^2$

Also,

$\frac{dM}{dt} = \frac{dM}{dt}\cdot\frac{dl}{dl} = \frac{dM}{dl}\cdot\frac{dl}{dt} = mv$

$\frac{d(KE)}{dt} = \frac{1}{2}(mv)(v^2) = \frac{1}{2}mv^3$

28) **Question explanation:**
Find the work done on a body in 2 seconds given its displacement as a function of time.
Concept:
Work-Energy Theorem: Work done = Change in kinetic energy
Solution:
Displacement $x = \frac{t^3}{3}$
Velocity $v = \frac{dx}{dt} = \frac{3t^2}{3} = t^2$
Initial velocity at t = 0: $v_i = 0^2 = 0$ m/s
Final velocity at t = 2: $v_f = 2^2 = 4$ m/s
Mass m = 2kg
Work done $W = \Delta K = \frac{1}{2}mv_f^2 - \frac{1}{2}m\,v_i^2$
$W = \frac{1}{2} \times 2 \times (4^2 - 0)$
W = 16 J
Final Answer: Option(4)

29) let r = real expansion
$r = x + 3\alpha_{brass}$
$r = y + 3\alpha_{tin}$
$x + 3\alpha_{brass} = y + 3\alpha_{tin}$
$\alpha_{tin} = \frac{3\alpha - y + x}{3}$

30)

Explain question : For a uniform metal cylider $\frac{\Delta \ell}{\ell} \times 100 = 3\%$, we have to calculate $\frac{\Delta A}{A} \times 100 = ?$

Concept : This question is based on Thermal expansion in solids

Solution :- for uniform metallic cylinderb

$\beta = 2\alpha$

$\frac{\Delta \ell}{\ell} \times 100 = \alpha \Delta T = 3\%$

$\frac{\Delta A}{A} \times 100 = \beta \Delta T$

$\frac{\Delta A}{A} \times 100 = 2\alpha \Delta T$

$\frac{\Delta A}{A} \times 100 = 2 \times 3 = 6$

Final answer :- 4

31) **Explain Question:**

In this question, we are asked the final temperature when ice at 0°C is mixed with hot water.

Concept:

Principle of Calorimetry- Mixing of different phases

Solution:

Given:

• Ice mass = 50 g at 0°C

• Water mass = 50 g at 60°C

• $S_w = 4.2$ J/g°C

• $L_f = 334$ J/g

Heat released by water (cooling to 0°C):

$Q_w = ms\Delta T = 50 \times 4.2 \times 60 = 12600$ J

Heat required to melt entire ice:

$Q_i = mL_f = 50 \times 334 = 16700$ J

Since

$Q_w < Q_i$

water does not have enough heat to melt all the ice, so mixture stays at 0°C.

Final Answer: Option (2)

32)

Explain the Question : Water has maximum density at 4°C. A beaker completely filled at 4°C will overflow if its temperature changes in either direction (heating or cooling).

Concept : This question is based on anomalous expansion of water : Density decreases (Volume increases) for Temperature above or below 4°C

Solution :

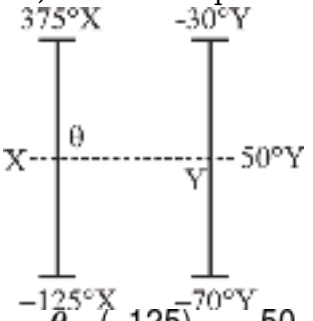
At 4°C, water has minimum volume (max density)

If heated → expands (since above 4°C volume increases) → overflows

If cooled → expands (since below 4°C volume increases again) → overflows

So overflow happens in both cases.
Final Answer : (3)

33) Let θ = temperature on X–scale corresponding to 50° on Y–scale



$$\Rightarrow \frac{\theta-(-125)}{375-(-125)}=\frac{50-(-70)}{-30-(-70)}$$

⇒ θ = 1375°X

34)

Question Explanation : A crystal has different coefficients of linear expansion along different directions. One principal direction has coefficient α_1, while the other two mutually perpendicular directions have coefficient α_2. We must find the coefficient of volume expansion.

Concept : This question is based on coefficient of volume expansion

Solution : The coefficient of volume expansion equals the sum of linear expansion coefficients along three perpendicular directions.
For small temperature change,
$\gamma = \alpha_x + \alpha_y + \alpha_z$
Here, $\alpha_x = \alpha_1$ and $\alpha_y = \alpha_z = \alpha_2$
So, $\gamma = \alpha_1 + 2\alpha_2$
Final Answer : Option (3)

35) Applying energy conservation :
$mgh = mL_f$

$$\Rightarrow h=\frac{L_f}{g}=\frac{3.36\times10^5}{10}$$

= 33.6 km

36) **Explain :**
A bimetallic strip, made of two metals with different expansion rates, is heated. determine how it will behave.
Concept : Bimetallic Strip
Formula : $\Delta L = \alpha L_0 \Delta T$
Calculation :
When heated, both metals in the bimetallic strip will expand. However, since they have different coefficients of linear expansion, they will expand by different amounts. The metal

with the higher coefficient of linear expansion will expand more. This differential expansion will cause the bimetallic strip to bend in the form of an arc, with the metal that expands more on the outer side of the arc (longer circumference).

Final Answer :

Option (3)

37)

$$\frac{F-32}{9}=\frac{K-273}{5} \Rightarrow \frac{x-32}{9}=\frac{x-273}{5}$$

$\Rightarrow$ x = 574.25

38) $T=2\pi\sqrt{\frac{\ell}{g}}$

$$\frac{\Delta T}{T_1}\times 100=\frac{1}{2}\times\frac{\Delta\ell}{\ell}\times 100$$
$$=\frac{1}{2}(\alpha\Delta t)\times 100$$
$$=\frac{1}{2}\times\left(2\times 10^{-6}\right)\times 10\times 100$$
$$=1\times 10^{-3}$$

39) Let final temperature of water be θ

Heat taken = Heat given

$\Rightarrow$ 110 × 1(θ – 10) + 10 (θ – 10)

= 220 × 1(72 – θ)

$\Rightarrow \theta \approx 50°C$

40) $L_t = L_0 (1 + \alpha t)$

$\Delta L = L_t - L_0 = L_0\alpha t$

If the length of each rod increases by same amount on raising the temperature to t, then

$\ell_1\alpha_1 t = \ell_2\alpha_2 t$

or $\ell_1\alpha_1 = \ell_2\alpha_2$

$$\frac{l_2}{l_1}=\frac{\alpha_1}{\alpha_2}$$

or

$$1+\frac{l_2}{l_1}=1+\frac{\alpha_1}{\alpha_2}$$
$$\frac{l_1+l_2}{l_1}=\frac{\alpha_1+\alpha_2}{\alpha_2}$$

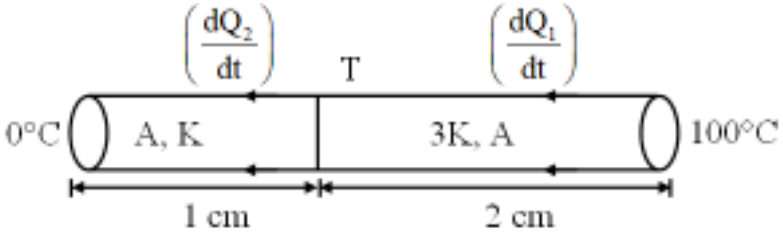
or $$\frac{l_1}{l_1+l_2}=\frac{\alpha_2}{\alpha_1+\alpha_2}$$

41) Temperature remain constant during phase conversion. Phase change at low temperature is freezing/melting point ie. 50°C and phase change at high temperature is boiling point ie. 200°C.

42) **Question Explanation :** We have to calculate temperature of the interface for given composite rod.

Concept : This question is based on heat current in steady state.

Solution :



In steady state

$H_1 = H_2$

$$\frac{dQ_1}{dt} = \frac{dQ_2}{dt}$$

$$\frac{3KA(100-T)}{2} = \frac{KA(T-0)}{1}$$

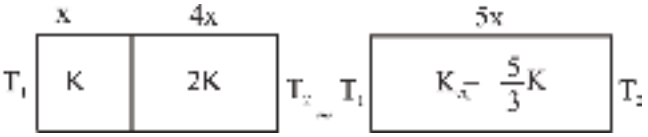
$300 - 3T = 2T$

$5T = 300$

$$T = \frac{300}{5} = 60°C$$

Final answer : option (3)

43)



$$K_{eq} = \frac{\ell_1 + \ell_2}{\frac{\ell_1}{K_1} + \frac{\ell_2}{K_2}}$$ (In series Combination)

Find ; $H = \frac{K_{eq}A}{\ell_{eq}}(T_2 - T_1)$ and compare with $\left[\frac{K(T_2 - T_1)A}{x}\right]$ f

we get $f = \frac{1}{3}$

44) $$K_{eq} = \frac{K_1 + K_2}{2} = \frac{K + 2K}{2} = \frac{3K}{2}$$

45) $T_B = 20$

$i_{DB} = i_{BC}$

$$\frac{(90-20)KA}{\ell_{BD}} = \frac{(20-0)KA}{\ell_{BC}}$$

$$\frac{\ell_{BD}}{\ell_{BC}} = \frac{7}{2}$$

CHEMISTRY

46) For all,

$K_{sp} = S^2$

If $K_{sp} \uparrow \Rightarrow S \uparrow$

47) **Explain Question:** Calculate the number of hydroxide ions present in 1 cc of solution having pH = 12.
Concept: Relation between pH and pOH
Solution: Given pH = 12, pOH = 14 − 12 = 2
So, $[OH^-] = 10^{-2}$ mol L^{-1}
Volume = 1 cc = 1 mL = 1×10^{-3} L
Moles of OH^- = concentration × volume
= $10^{-2} \times 10^{-3}$
= 10^{-5} mol
Number of OH^- ions = moles × Avogadro number
= $10^{-5} \times 6.022 \times 10^{23}$
= 6.022×10^{18} ions
Final Answer: Option (4)

48) $pOH = pK_b + \log \frac{[salt]}{[base]} = 5 - \log(1.8) + \log\left(\frac{0.25}{0.15}\right)$
= 5 – log 1.8 + log 5 – log 3
pOH = 4.96
So, pH = 14 – 4.96 = 9.04

49) **(A) $Al(OH)_3$**
$K_{sp} = (1)^1(3)^3S^{1+3}$
$K_{sp} = 27\ S^4 \Rightarrow S = \left(\frac{K_{sp}}{27}\right)^{1/4}$
(B) $BaCrO_4$
$K_{sp} = S^2 \Rightarrow S = (K_{sp})^{1/2}$
(C) $Zr_3(PO_4)_4$
$K_{sp} = (3)^3(4)^4S^7$
$S = \left(\frac{K_{sp}}{6917}\right)^{1/7}$
(D) Hg_2I_2
$K_{sp} = 4\ S^3 \Rightarrow S = \left(\frac{K_{sp}}{4}\right)^{1/3}$

50)

$EQ_{H_2SO_4} = 10^{-1} \times 10 = 1$
$EQ_{NaOH} = 10^{-1} \times 10 = 1$
So, neutral Solution pH = 7

51)

Question Explanation:

Tests the calculation of pH for strong acids and strong bases considering water ionization.
Concept Used:
pH Scale and Strong Acid–Strong Base
Answer Explanation:

A. (A) 10^{-2} M HCl → $[H^+] = 10^{-2}$ → pH = 2 → (ii)
B. (B) 10^{-2} M NaOH → $[OH^-] = 10^{-2}$ → pOH = 2, pH = 12 → (i)
C. (C) 10^{-7} M HCl → Neutral water effect considered → pH ≈ 6.8 → (iv)
D. (D) 10^{-8} M NaOH → Water ionization dominates → pH ≈ 7.1 → (iii)

Final Answer:
Option (3)

52) NH_4^+ is not an amphoteric species.

53) H_2CO_3/HCO_3^- is correct answer.

54) **Question Explanation:** The question asks for the solubility of $PbSO_4$ in a common ion solution (0.01 M Na_2SO_4).

Concept: The common ion effect decreases the solubility of a salt. For $PbSO_4$, Ksp = $[Pb^{2+}][SO_4^{2-}]$.
In 0.01 M Na_2SO_4, $[SO_4^{2-}]$ = 0.01 M is already present.

Solution:
Let the solubility of $PbSO_4$ be s (in mol/L). Then:
Ksp = $[Pb^{2+}][SO_4^{2-}]$ = s × (0.01)
1.25×10^{-9} = s × 0.01
s = 1.25×10^{-9} / 0.01 = 1.25×10^{-7} mol L^{-1}

Final Answer: Option (1)

55)

$K_a = C \cdot \alpha^2$

$1.8 \times 10^{-5} = C \times \left(\frac{2}{100}\right)^2$
C = 0.045 M

56) First dissociation is more than second dissociation.

57)

Explain Question:
The question is asking: If the pH of pure of water is 6.63, what does this indicate about the temperature of Water?

Concept:
For pure water: $[H^+] = [OH^-]$. At 298 K, $K_w = 1 \times 10^{-14}$, so pH = 7. With rise in temperature, K_w increases, $[H^+]$ increases, and pH decreases below 7.

Solution:
* At 298 K, pH = 7.
* Given pH = 6.63 (<7), this means $[H^+]$is higher due to increased K_w.
* Hence, temperature must be greater than 298 K.

Final answer: Option (3)

58) $\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightarrow NH_3; \Delta H = -46kJ/mol$
 for $NH_3 \rightarrow \frac{1}{2}N_2 + \frac{3}{2}H_2; \Delta H = +46kJ/mol$
and for $2NH_3 \rightarrow N_2 + 3H_2$; $\Delta H = +92$ kJ/mol

59) **Question Explanation :** We need to find ΔH of formation of $N_2H_4(g)$ using given values of bond energy.

Concept : $\Delta H \sum(B.E)_R - \sum(B.E)_P$

Solution :
$N_2 + 2H_2 \rightarrow N_2H_4$
$\Delta H = [(B.E)_{N\equiv N} + 2(B.E)_{H-H}] - [(B.E)_{N-N} + 4\ (B.E)_{N-H}]$
$\Delta H = [941 + 2\ (436)] - [159 + 4\ (398)]$
$\Delta H = 62$ kJ/mol

Final Answer : Option (2)

60) $\frac{1}{2}X_2 + \frac{1}{2}Y_2 \rightarrow XY$
$\Delta H_f^\circ = \Sigma BE_R = \Sigma BE_P$
$-200 = \left[\frac{1}{2}\Delta H_{X-X}^\circ + \frac{1}{2}\Delta H_{Y-Y}^\circ\right] - \left[\Delta H_{X-Y}^\circ\right]$
$-200 = \left(\frac{x}{2} + \frac{x}{4}\right) - x$
$-200 = \frac{-x}{4}$
x = 800

61) **Explain Question:**
Using heats of combustion of C_2H_4, C_2H_6 and H_2 determine the enthalpy change for hydrogenation reaction forming ethane.

Concept:
$\Delta H_{rxn} = \sum \Delta H_c(\text{reactants}) - \sum \Delta H_c(\text{products})$

Solution:

Hydrogenation reaction: $C_2H_4 + H_2 \rightarrow C_2H_6$ **Using combustion formula:** $\Delta H_{rxn} = \Delta H_c(C_2H_4) + \Delta H_c(H_2) - \Delta H_c(C_2H_6)$ **Substitute values:** $\Delta H_{rxn} = (-1409.5) + (-285.6) - (-1558.3)$
$\Delta H_{rxn} = -1695.1 + 1558.3$
$\Delta H_{rxn} = -136.8$ kJ
Thus the heat of hydrogenation of ethene is -136.8 kJ.

Final Answer: Option (1)

62) **Question Explanation:** Determine heat of atomization of hydrogen molecule from given enthalpy change for dissociation of H_2 into hydrogen atoms.
Concept: Heat of Atomisation.
Solution:
Heat of atomisation is the enthalpy required to convert **one mole of a substance into its gaseous atoms**.
Given reaction:
$H_2(g) \rightarrow 2H(g) \quad \Delta H = 104$ kcal
This reaction already represents the complete atomisation of 1 mole of H_2 molecules into gaseous hydrogen atoms. Therefore the enthalpy change of this reaction itself is the heat of atomisation of hydrogen molecule.
Heat of atomisation = 104 kcal
Final Answer: Option (2)

63)

$C_{(graphite)} + O_{2(g)} \rightarrow CO_{2(g)}$
$\Delta H = (\Delta H_f)_{CO_2} = (\Delta H_C)_{(graphite)}$

64) $2C\ (graphite) + 3H_2(g) \rightarrow C_2H_6\ (g)$

65)

$n = \frac{5.8}{58} = 0.1$ mol
Now, for
2 mol → 5756 KJ
0.1 mol → $\frac{5756}{2} \times 0.1 = 287.8$ KJ

66)

$NaCl + H_2O \rightarrow NaOH + HCl \quad \Delta H = 55.9$ kJ/mol ....(1)
$HCN + NaOH \rightarrow NaCN + H_2O \quad \Delta H = -12.1$ kJ/mol ....(2)
Add eq (1) & (2)
$Na^+Cl^- + HCN \rightarrow Na^+CN^- + H^+Cl^-$
$\Delta H_{net} = -12.1 + 55.9 = 43.8$ kJ/mol
net reaction $HCN \rightarrow H^+ + CN^-$

67)

Explanation :- Find ΔG° in kJ
Keq = 10, T = 300 K

Concept :- This question is based on Relation between ΔG° and Keq

Solution :- ΔG° = -2.303 RT log Keq ; $R = \left(\frac{8.314}{1000}\right)\frac{kJ}{mol\ k}$

$= -2.303\left(\frac{8.314}{1000}\right)(300)$ log 10

$= -2.303\left(\frac{8.314}{1000}\right)(300)$

= -

= - 5.744 kJ

Answer :- 1

68)

We know, ΔG° = – 2.303 RT $\log_{10}$ Kc,

R = 2 cal K^{-1} mol^{-1}, T = 300 K,

$K_c = \frac{10 \times 15}{3 \times 5} = 10$

ΔG° = – 2.303 × 2 × 300 $\log_{10}10$ = – 1381.8 cal

69) **Question Explanaiton:**
Identify the reaction that is never spontaneous under any conditions.

Concept:
This question is based on ΔG = ΔH – TΔS

Solution:
For a reaction to never be spontaneous ΔH = + ve and ΔS = – ve (ΔG always > O) at any temperature.

Only opton (4) has ΔH = +ve, ΔS = – ve, ΔG = + ve

Correct Option: 4

70) If ΔG° > 0 ⇒ K < 1

71) Oxidation number of oxygen is –2 in most of its compounds. The exceptions are –1 in peroxides and $-\frac{1}{2}$ in superoxides, +2 in OF_2 and +1 in O_2F_2.

72) $Na_2\overset{a}{S_4}O_6 : 2 + 4a + 6 \times (-2) = 0$

$$a = +\frac{5}{2} = +2.5$$

73) $KO_3 \Rightarrow O = -1/3$,
$Na_2O_2 \Rightarrow O = -1$

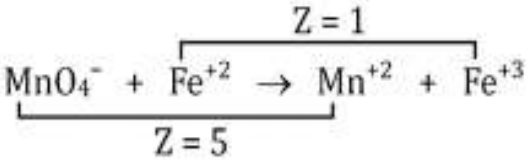
74) n-factor = $|2 \times 2 + 8 \times 3| \Rightarrow 28$

75) Total +ve and total –ve must be same
$A_3(B\ C_4)_2$
Total +ve = 16
Total –ve = 16

76) $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{+2} + 4H_2O$
so n = 5.

77)

After balancing→
$4Zn + NO_3^- + 10H^+ \rightarrow 4Zn^{+2} + NH_4^+ + 3H_2O$
Coefficient of Zn = 4
Coefficient of NO_3^- = 1
Coefficient of H^+ = 10

78)


$MnO_4^- + 5Fe^{+2} \rightarrow Mn^{+2} + 5Fe^{+3}$

79) **Question Explanation :** We are asked to identify which reaction is a disproportionation reaction.

Concept :
Disproportionation

Solution :

A. Disproportionation involves the same element undergoing both oxidation and reduction.
B. In option 3:

$I_2 + OH^- \rightarrow I^- + IO^- + H_2O$ Iodine is reduced to I^- and oxidized to IO^- classic disproportionation.

A. Other options involve redox between different species, not self-oxidation/reduction.

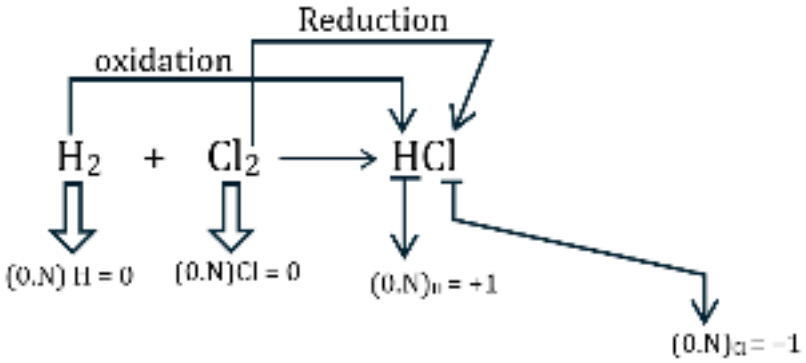
Final Answer : Option (3)

80)

Question Explanation : We need to find which reaction is redox reaction.

Concept : This question is based on Redox Reaction.

Solution :



Reaction is Redox Reaction.

Final Answer: option (3)

81) NCERT Pg-131
⇒ HF > HI > HBr > HCl

82) NCERT Pg-132

83) NCERT Pg-303

84) NCERT Pg-106

85) NCERT Pg-111

86) NCERT Pg-106
NaCl being ionic has high melting point.

87) Due to back bonding

88)

Dimer of $BeCl_2$ has 3c-4e$^-$ bond.

89) $Mg_3N_2 + H_2O \rightarrow Mg(OH)_2 + NH_3$

90) Boron trioxide being non-metal oxide is acidic.

BIOLOGY

91) NCERT (XIth) Pg.#145 and 148

92) NCERT XI Pg. # 146

93) NCERT Pg. # 145

94) **Explain Question :** The question asks to identify which of the given options is a lipid found in the cell membrane

Concept : Conjugated And Derived Lipid

Explanation : Phospholipids like lecithin form the bilayer structure of cell membranes.

Final answer : 3

95) NCERT Pg. # 145

96)

NCERT, Pg. # 106

97)

From Module Pg. # 10

98)

Module Pg. # 4

99)

Explain Question : The question asks to classify the molecule Lecithin into one of the provided categories of lipids.

Concept : Conjugated And Derived Lipid

Explanation : “Phospholipids, such as lecithin, are components of the cell membrane.”

Final answer : 2

100) NCERT XI Pg. # 147 last para

101)

From Module Pg. # 5

102)

Module

103) NCERT (XIth) Pg. # 146

104)

Question Explanation: select odd ones with respect to primary metabolites

Concept: primary and secondary metabolites

Solution: drugs are a part of secondary metabolites not primary metabolites

Final answer: 3

105)

Question Explanation: Bond present between amino acids in protein

Concept: configuration of protein

Solution: In proteins, amino acids are linked together by peptide bonds, which are a type of amide bond.

A. Peptide bond: This is the specific name for the bond formed between the carboxyl group (-COOH) of one amino acid and the amino group (-NH2) of another amino acid.

B. Amide bond: This is a general term for a bond formed between a carbonyl group (C=O) and an amine group (-NH2). A peptide bond is a specific type of amide bond.

Final answer: Peptide bond and Amide bond.

106)

Explain Question:
This question is asking about the defining characteristics of α-amino acids.

Concept:
Biomolecules – Structure of amino acids.

Solution:
In a-amino acids, the amino and carboxyl groups are attached the same a-carbon; there are 20 standard a-amino acids that participate in protein synthesis.

Final answer: option (4)

107)

Explanation:

Aromatic amino acids contain a benzene or similar ring in their side chain. The three aromatic amino acids are:

A. Tyrosine (Option 1): Contains a phenol (-OH attached to a benzene ring).

B. Tryptophan (Option 2): Has an indole ring (a fused double-ring structure).

C. Phenylalanine (Option 3): Contains a benzyl group ($-CH_2-C_6H_6$)

Correct Answer: 4. All the above

108) NCERT Pg.#144

109)

From Module

110)

From Module

111)

NCERT, Pg. # 107

112)

NCERT Pg. # 148 - 149

113)

NCERT, Pg. # 107

114) NCERT XI, Pg. # 150

115)

NCERT, Pg. # 106

116)

NCERT, Pg. # 109

117)

From Module

118)

From Module

119)

NCERT, Pg. # 109

120) **Solution:**
Reason: The correct structure of an **acidic amino acid** (Aspartic acid or Glutamic acid) must have an **extra -COOH group** in the side chain
The Correct answer is: 2. C

121)

NCERT, Pg. # 109

122) NCERT-XI, Pg # 145, Fig. 9.1

123) NCERT-XI, Pg. # 147

124)

NCERT-XI, Pg. # 157

125)

NCERT-XI, Pg. # 157

126) NCERT-XI, Pg. # 163

127)

NCERT-XI, Pg. # 153, 154

128)

NCERT-XI, Pg. # 159,160

129)

NCERT-XI, Pg. # 160

130)

NCERT-XI, Pg. # 158, 159

131)

NCERT-XI, Pg. # 139, 140

132)

NCERT-XI, Pg. # 149

133)

NCERT-XI, Pg. # 157,158,159

134) NCERT-XI, Pg. # 144

135)

NCERT-XI, Pg. # 149

136)

NCERT-XI, Pg. # 145, 146

137)

NCERT-XI, Pg. # 145, 146

138)

NCERT-XI, Pg. # 143, 144

139)

NCERT-XI, Pg. # 138, 139

140)

NCERT-XI, Pg. # 136, 137

141)

NCERT-XI, Pg. # 136

142)

NCERT-XI, Pg. # 138, 139

143)

NCERT-XI, Pg. # 159

144)

NCERT-XI, Pg. # 140, 141

145)

NCERT-XI, Pg. # 140, 141

146)

NCERT-XI, Pg. # 150

147)

NCERT-XI, Pg. # 163, 164

148)

NCERT-XI, Pg. # 161, 162

149)

NCERT-XI, Pg. # 157

150)

NCERT-XI, Pg. # 158, 159

151)

NCERT-XI, Pg. # 160, 161

152)

NCERT-XI, Pg. # 158, 159

153)

NCERT-XI, Pg. # 30, 31

154)

NCERT-XI, Pg. # 32, 33

155)

NCERT-XI, Pg. # 28, 29

156)

NCERT-XI, Pg. # 25, 26

157) NCERT-XI, Pg. # 24 to 32

158) NCERT XI, Pg. # 24

159) NCERT-XI, Pg. # 24

160) NCERT-XI, Pg. # 28, 29

161)

NCERT-XI, Pg. # 28

162) NCERT-XI, Pg. # 32

163)

NCERT-XI, Pg. # 29

164) NCERT-XI, Pg. # 11, 12

165)

NCERT XI, Pg. # 16 to 18

166) NCERT-XI, Pg. # 16

167) NCERT-XI, Pg # 13

168)

NCERT-XI, Pg. # 13

169)

NCERT-XI, Pg. # 15

170)

NCERT-XI, Pg. # 16, 17

171) NCERT XI, Pg # 17

172)

NCERT-XI, Pg. # 17, 18

173) NCERT-XI, Pg. # 19

174) NCERT-XI, Pg. # 12, 13

175)

NCERT-XI, Pg. # 3

176) NCERT, XI Pg. # 3

177)

NCERT-XI, Pg. # 143, 144

178)

NCERT-XI, Pg. # 143, 144

179) NCERT-XI, Pg. # 143

180)

NCERT-XI, Pg. # 149,150